Supervised machine learning reveals high efficacy of mobile elements to discriminate *Salmonella* outbreaks

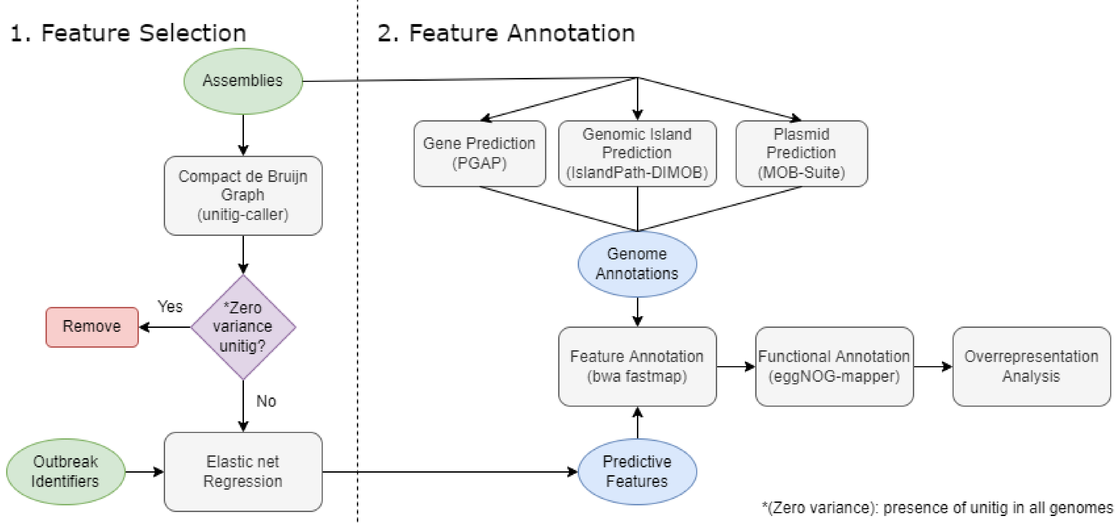
Chao Chun Liu[1] and William Hsiao[1,2]

[1] Simon Fraser University, Burnaby, Canada; [2] University of British Columbia, Vancouver, Canada

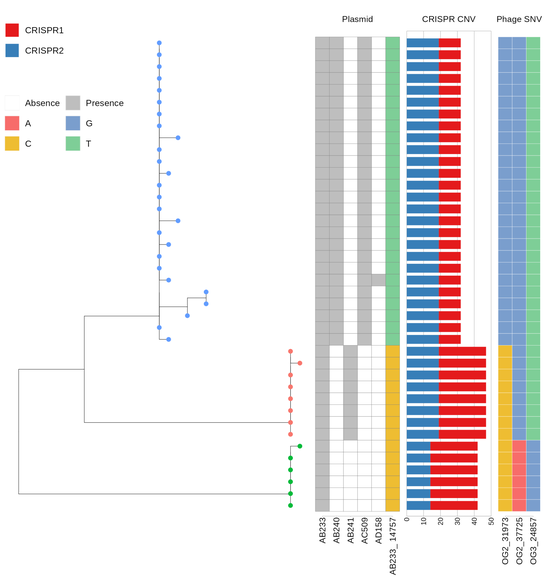
Background

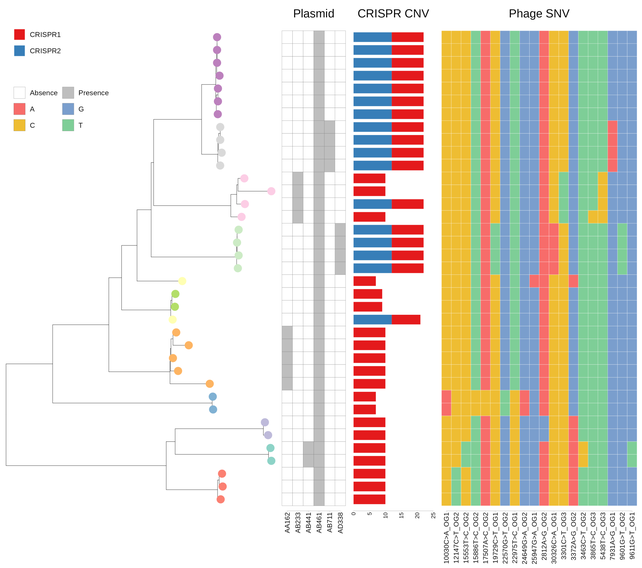
The high-resolution typing of pathogens empowered by genomics in conjunction with the contextualization of diseases by epidemiology have contributed to great successes in identifying outbreak origins and linking related cases. Existing methods to infer clonal strains in surveillance settings primarily involve the comparison of genetic variation in a stable segment of pathogen genomes known as the core genome. In consequence, the non-conserved genetic elements are often neglected and contribute no information to the genetic distance between two query genomes. To demonstrate the analytical value of non-conserved elements to link outbreak related cases, our study conducted regularized regression modeling on 24 historical foodborne outbreaks caused by *Salmonella enterica* to characterize genetic features predictive of outbreak clusters.

Analysis Workflow

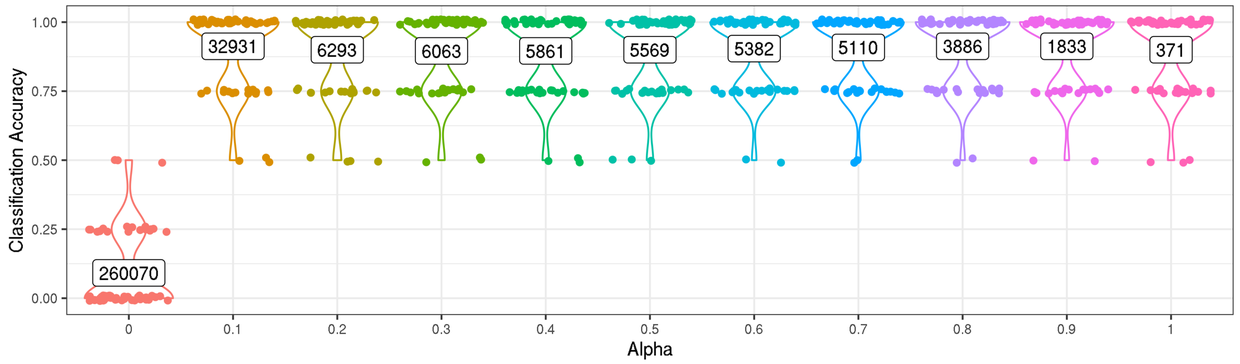


Explaining the Discriminatory Power of Mobile Elements





Model Performance Summary



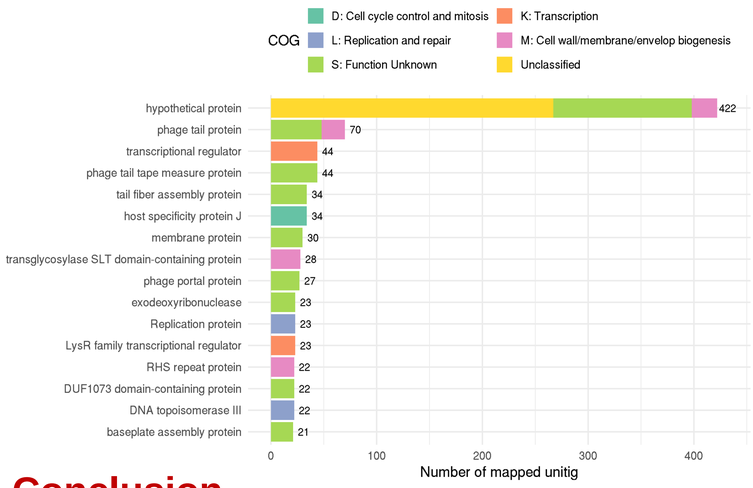
Impact of alpha value (ratio of L1 and L2 regularization) on regularized regression model performance. The numeric text label and its relative y-axis position at each alpha value summarize the total number of non-zero coefficient predictors and overall prediction performance averaged across cross-validation folds (k = 60), respectively.

Feature Annotations

Functional annotations of unitigs predictive of *Salmonella* outbreak clusters

	Chromosome			Extrachromosome	
	GI[1]	Phage	Other	Plasmid	Phage
CDS[2]	52	772	3810	440	117
IGR[3]	5	33	450	74	—
pseudogene	8	9	173	28	2
CRISPR Repeat	—	—	26	—	—
tRNA	—	—	16	—	—
Putative CDS	—	—	6	—	—
rRNA	—	—	5	—	—
tmRNA	—	—	2	—	—
ncRNA	—	—	1	—	—
Total	65	814	4,489	542	119

[1] Genomic Island
[2] Coding sequence
[3] Intergenic region



Conclusion

- A significant proportion of outbreak predictors were found to map to accessory elements that contribute to large indels
- The outbreak predictors represent a set of highly mutable sites whose short-term evolutionary trajectories are likely driven by selection or the lack of functional constraints
- Existing *Salmonella* outbreak cluster detection methods can benefit from the additional comparisons of genetic variations in the hypervariable elements identified in our study
- The proposed analytical framework can be readily applied to outbreaks caused by other infectious disease agents to investigate inter-species differences of predictive features

Acknowledgement

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Partners

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